

ME 621: Aircraft Propulsion

Course Code	ME 621
Course Name	Aircraft Propulsion
L-T-P-C	2-1-0-3
Prerequisites	Thermodynamics
Intended for	UG/MS/PhD
Distribution	Elective
Approval	6th Senate (May 2014)

Course Contents

- **Module I**
Aircraft propulsive devices / engines (basic principle of operation). [3 Lectures]
- **Module II**
Brief information on - piston-prop, turbo-prop, turbojet, turbofan, turbo-shaft, ramjet, scramjet, vectored thrust (incl. lifting engines)
Aircraft gas turbine engine vs. Industrial gas turbine engine (contrast). [1 Lectures]
- **Module III**
Thermodynamic, aerodynamic, & mechanical design considerations in aircraft gas turbine engine. [3 Lectures]
- **Module IV**
Aircraft propulsion mechanics, concepts, & performance measures. [2 Lectures]
Thrust (T), Specific Thrust (F_{sp}), Propulsive Efficiency, Overall efficiency, Thrust specific fuel consumption.
- **Gas turbine cycle (Joule) for Aircraft / Helicopter propulsion**
Open cycle single shaft and twin shaft arrangements, multispool arrangements. [2 Lectures]
- **Thermodynamic analysis of ideal shaft power cycles**
T-S diagram, cycle efficiency, specific work output (derivations). [3 Lectures]
- **Thermodynamic analysis of ideal reheat cycle**
T-S diagram, optimum point of reheat, cycle efficiency & specific work output (derivations)
Thrust augmentation using after burner and water injection in compressor inlet.
- **Methods for accounting for component losses (with illustrations on T-S diagram)**
Compressor and Turbine efficiencies [isentropic and polytropic (including incides for compression and expansion)], intake & propelling nozzle efficiencies, mechanical losses
Fuel air ratio, combustion efficiency (& relation with thermal efficiency) [4 Lectures]
- **Simple Turbo-Jet Engine (TJE) cycle**
Calculation of F , F_{sp} , η_{pr} , sfc;
Variation of F , F_{sp} , η_{pr} , sfc with flight conditions for given TJE
- **Analysis of flow through - compressor, turbine, combustion chamber, afterburner, nozzle** [2 Lectures]
- **Variable area nozzle, thrust spoiler & reverser, engine noise suppressor** [2 Lectures]
- **Bypass engines**
Turbo-fan Engine (TFE) configurations, thermodynamic design point performance prediction of TFE, turbo-prop engine & its propeller efficiency
Optimization of TFE cycle for minizing sfc and maximizing F_{sp} based on bypass ratio and fan pressure ratio. [6 Lectures]
- **Off-design performance prediction of aircraft gas turbine engines based on component characteristics**
Relations based on work, flow, rpm compatibilities between components
Single shaft engine delivering shaft power, free turbine engine (incl- matching of two turbines in series, study of variation of power output & sfc with rpm of free turbine)
Jet engine (incl - matching of gas-generator turbine with nozzle, study of variation of F with engine rpm, forward speed, altitude)

Text Books

1. H.I.H. Saravanamuttoo, G.F.C. Rogers, H. Cohen, P. Straznicky, **Gas Turbine Theory**, 6th Edition, Prentice Hall / Pearson Education, Singapore, 2009.
2. P.G. Hill, C.R. Peterson, **Mechanics and Thermodynamics of Propulsion**, 2nd Edition, Addison-Wesley Co. Reading MA USA, 1992.
3. R.D. Flack, **Fundamentals of Jet Propulsion with Applications**, Cambridge University Press, Cambridge, 2005.